Original Article

Revolutionizing Type 2 Diabetes Care: Scientific Journey of Fiber Rich Cookies Made from Green Unripe Jackfruit and Sorghum Flour

Rakesh U. Shelke[1] | Anjali D. Kshirsagar[1] | Tanishk D. Khairnar[1] | Payal V. Patil[1] | Eknath D. Ahire[2] | Khemchand R. Surana[3,*]

[1]Department of Pharmaceutical Analysis, MET Institute of Pharmacy, Adgaon, Nashik, Affiliated to Savitribai Phule Pune University, Pune, Maharashtra 422003, India
[2]Department of Pharmaceutical Quality Assurance, MET Institute of Pharmacy, Adgaon, Nashik, Affiliated to Savitribai Phule Pune University, Pune, Maharashtra 422003, India
[3]Department of Pharmaceutical Chemistry, SSS's, Divine College of Pharmacy, Satana, Nashik, Affiliated to Savitribai Phule Pune University, Pune, Maharashtra 423301, India

Use your device to scan and read the article online

Citation R.U. Shelke, A.D. Kshirsagar, T.D. Khairnar, P.V. Patil, E.D. Ahire, K.R. Surana, **Revolutionizing Type 2 Diabetes Care: Scientific Journey of Fiber Rich Cookies Made from Green Unripe Jackfruit and Sorghum Flour.** *Eurasian J. Sci. Technol.*, **2026**, 2026, 6(1), 84-95.

https://doi.org/10.48309/ejst.2026.556765.1281

CC BY

Article info:
Received: 2025-10-01
Accepted: 2025-11-07
Available Online: 2025-12-28
ID: EJST-2511-1281

Keywords:
Diabetes, Antidiabetic, Cookies, Unripe jackfruit, Blood glucose levels, Nutrition.

ABSTRACT

Artocarpus heterophyllus (jackfruit) used in the formulation is the best way to control or maintain blood glucose levels. In small intestine, the high levels of fiber in jackfruit slow digestion by interacting with sugar, cholesterol, fat, and decrease their absorption in the body. The formulated supplement mainly contains gluten free sorghum flour and date syrup which is rich source of vitamin B complex along with seeds that have antioxidant properties. These ingredients also help to improve the nutritional efficacy of the product, prevent heart diseases, and maintain blood glucose levels. Together, these components minimize the quick absorption of sugar and offer a balance of macronutrients, making your cookies a healthier choice for diabetics. Evaluated parameters include general appearance, qualitative tests, patient acceptability, HbA1c tests (blood glucose level tests), and more. Additionally, these cookies exhibit a lower glycemic impact compared to traditional ones, making them a more desirable option. The inclusion of green unripe jackfruit in the formulation of cookies results in low glycemic index, which could potentially, provide health benefits for individuals suffering from degenerative diseases such as diabetes mellitus. The incorporation of functional ingredients like green unripe jackfruit, dates, and sorghum in the development of novel products holds great promise for the food industry. From this study it can be concluded that, this approach not only helps reduce waste, but also presents opportunities for cost reduction in the production of baked goods. The use of unripe jackfruit in cookie manufacturing addresses both environmental concerns and economic efficiency in the baking industry.

Introduction

Diabetes mellitus is a non-communicable, inherited metabolic disorder characterized by high blood sugar and abnormal carbohydrate metabolism. It is leading of morbidity and mortality, affecting nearly 9% of global adult population. It is also associated with other diseases, which can either result from decreased

*Corresponding Author: Khemchand R. Surana: khemchandsurana772@gmail.com

insulin secretion by the beta cells of pancreatic Langerhans or from increased insulin resistance due to excessive glucose absorption [1].

The main stages of the treatment of the diabetes are synthetic oral hypoglycemic drugs and insulin. Since these drugs have eminent side effects and are unable to reverse the course of their complications, the focus has been shifted to formulating anti-diabetic cookies which can be part of a healthy meal plan and also provide quick energy to diabetic patients [2,3]. Notwithstanding the availability of anti-diabetic drugs in the modern age, the treatment of diabetes with unripe flour from jackfruit and gluten free sorghum flour is a wholly novel idea [4].

However, in comparison to conventional drugs unripe jackfruit is rich source of fiber and possesses no side effects. Unripe jackfruit and gluten free sorghum are relatively inexpensive and widely available. Green unripe jack fruit and sorghum are nutrient-dense food sources [5,6]. Green unripe jackfruit has 40%fewer-carbohyadrates, 40% fewer calories and 40% lower glycemic index than wheat and rice, marking it a suitable option for individuals with dietary restrictions. Furthermore, it has 4 times more dietary fiber and contains antioxidant properties [7]. Conventional grains can be replaced by green jackfruit flour providing healthier anti-diabetic cookies. Green unripe jackfruit has four times more dietary fiber, which slows down the absorption of glucose [8]. Ozempic, a medication containing semaglutide, is utilized in the management of type 2 diabetes [9]. Semaglutide functions as both an anti-diabetic and anti-obesity drug, aiding in long-term weight control [10]. This peptide is structurally similar to the hormone glucagon-like peptide-1 (GLP-1) [11,12], but features a modified side chain. Jackfruit flour, rich in fiber, may offer comparable benefits in diabetes management. The fiber content in jackfruit powder is believed to function in a manner analogous to semaglutide [13].

It contains a significant amount of soluble dietary fibers, specifically pectin, which serves as nourishment for gut microorganisms. This fiber is then fermented and converted into short-chain fatty acids. The taste of green unripe jackfruit is neutral, making it a suitable option for individuals with diabetes [14]. Additionally, it has low bulk density, as indicated by the ratio of 30 g of rice flour to 50 g of jackfruit flour. Green unripe jackfruit is packed with a variety of vitamins and minerals [15], including vitamin A (carotenoids), thiamine, riboflavin, niacin, vitamin C (ascorbic acid) , and vitamin E (alpha tocopherol), as well as potassium, calcium, phosphorus, sodium, iron, and magnesium. Additionally, green unripe jackfruit flour is combined with gluten free sorghum flour to create a nutrient-rich, gluten free product. To make anti-diabetic cookies, date syrup, ghee, peanut butter, various seeds and other ingredients are also added to the dough [16]. Despite numerous studies exploring the potential of jackfruit-derived bioactive compounds in managing diabetes, few studies have focused on developing food products such as cookies using green unripe jackfruit flour as an ingredient [17].

Cornstarch, a common thickening agent for various dishes including soups, stews, sauces, and desserts, is rich in calories and carbohydrates but lacks essential nutrients such as protein, fiber, vitamins, and minerals. Its low fiber content results in rapid digestion [18], which can cause blood sugar levels to rise quickly, as fiber typically slows down sugar absorption into the bloodstream. Consequently, individuals with diabetes may want to reconsider incorporating cornstarch into their diet due to its potential to cause blood sugar spikes. A superior alternative to cornstarch is jackfruit flour [19], which is high in fiber content. Jackfruit flour can be utilized in various dishes, including Chinese cuisine, potentially offering health benefits [20]. In addition, the food industry currently faces the challenge of producing nutraceuticals and functional foods that are high in bioactive ingredients to meet the demand for these products. With these requirements in mind, green unripe jackfruit flour, which is higher in fiber and gluten free sorghum flour may be considered beneficial ingredients. As a result, this study was designed to develop a diabetic-friendly cookie by incorporating these ingredients and other nutritious components, and to evaluate its organoleptic properties, proximate analysis,

glycemic index, qualitative analysis and anti-diabetic properties.

Materials and Methods

Materials

Green unripe jackfruit was harvested from the local region of Nashik, Maharashtra, India. Sorghum flour, Dates, ghee, sunflower seeds, sesame seeds, almonds, peanut butter, baking soda, baking powder, and vanilla essence were purchased from authorized vendors in Nashik, Maharashtra, India. Table 1 represents the list of ingredients used in the preparation of cookies.

Table 1. List of ingredients

Sr./ No.	Ingredients	Quantity	Properties
1	Sorghum	70 g	Gluten free
2	Date syrup	10 mL	Iron source
3	Peanut butter	5 g	Protein source
4	Ghee	5 mL	Fatty acids
5	Sunflower seeds	Q.S	Vitamin rich
6	Almond	Q.S	Low calories
7	Sesame seed	Q.S	Vitamin rich
8	Vanilla essence	Q.S	Flavor

Preparation of Materials

Preparation of sorghum flour was obtained by cleaning to remove dirt, stones, and other extraneous materials. It is then dry milled, sieved, and packaged. Date syrup was prepared using the dates that were chopped and pitted before being heated in water. The mixture was then blended by pressing it through a filter to strain out pits and insoluble parts of the date. Afterwards, the water was evaporated and the resulting syrup was ready for use.

Dough Preparation

Commenced by acquiring a bowl and amalgamating peanut butter, ghee, date syrup and vanilla extract to form a creamed mixture. In a separate receptacle, green unripe jackfruit flour, sorghum flour, and baking powder were combined, and then it was incorporated into the creamed mixture. Subsequently, almonds, sesame seeds, and sunflower seeds were included, thoroughly mixed, and the ingredients were kneaded to create dough.

Preparation of Cookies

The oven was preheated to 180 ˚C and parchment paper was placed on the baking sheets. The dough was then taken, weighed and rolled properly. It was cut into uniform shapes using a cookie cutter and placed on the prepared baking sheets. The baking process was carried out in the preheated oven until the edges turned golden, which took approximately 20 to 25 minutes at 180 ˚C. After cooling, the baked cookies were packed and stored in a cool, dry place. Subsequently, various analysis methods were employed to evaluate their nutritional value, taste, and quality.

Sensory Evaluation

The purpose of the sensory evaluation was to gauge the level of consumer acceptability for green unripe jackfruit anti-diabetic cookies, with a focus on evaluating their taste, texture, color, odor, and level of overall satisfaction. The cookies were presented to the panelists in a controlled setting, where they evaluated each sample according to a 7 point hedonic scale that assessed sweetness, texture, color, and overall appeal. Specific focus was given to the impact of these factors on the perception of anti-diabetic cookies [21,22].

Odor

A 7-point hedonic scale was employed to assess the samples aroma. Panelists evaluated the scent as sweet aromatic, providing a detailed insight into consumer preferences regarding fragrance attributes [21,22].

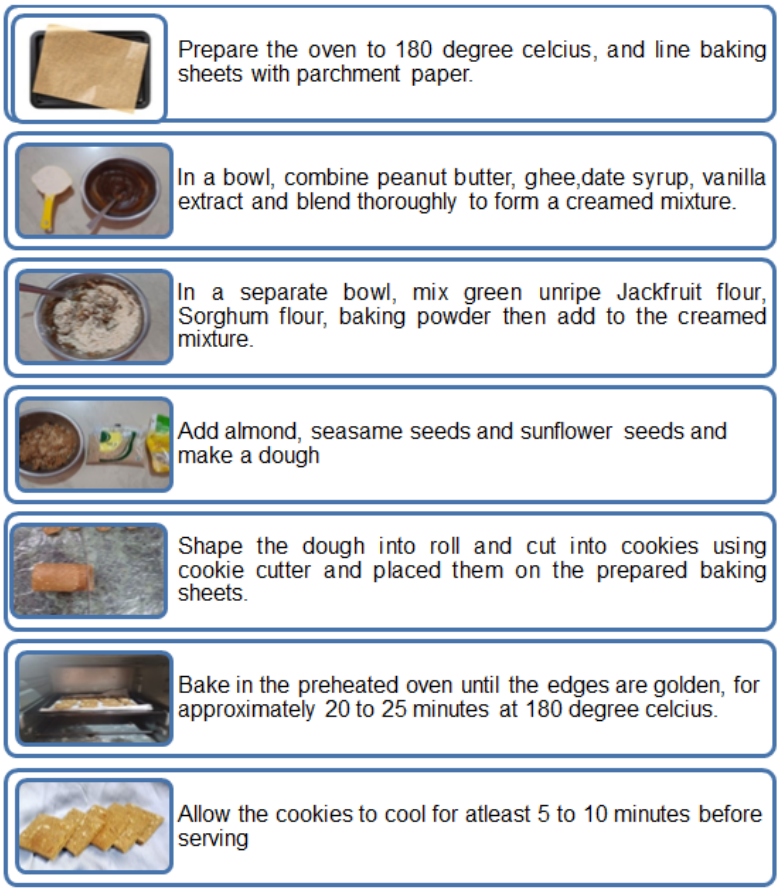


Figure 1. Process flow for the cookies preparation

Taste

Participants evaluated the flavor of each sample using a 7-point hedonic scale to measure their level of sensory satisfaction. The cookies were characterized as having a mildly sweet taste, which was determined to be the most desirable flavor profile [21,22].

Texture

The texture assessment was conducted utilizing a 7-point hedonic scale, allowing participants to indicate their preferences for mouthfeel characteristics. This approach provided insights into consumer acceptance. The cookies were found to have a crunchy texture with a coarse crumb [21, 22].

Color

Respondents rated the color of the samples on a 7-point hedonic scale, enabling an evaluation of the cookies' visual appeal. The color was described as brownish [21, 22].

Gluten Test

Gluten is a category of proteins that may be found in wheat, barley, and rice. When combined with water, it forms a sticky and flexible network. Despite its prevalence in certain grains, consuming gluten may have determental effects on one's health [23]. On the other hand (Jowar) sorghum and unripe jackfruit flour are devoid of gluten, making them suitable choices for those who must avoid it [24].

Bulk Density

Jackfruit flour has a relatively low bulk density. It is worth noting that 30 g of rice flour can be compared to 50 g of jackfruit flour / powder. As an illustration, consider the case where 2 biscuits of wheat flour correspond to the only one biscuit of jackfruit flour [25].

Qualitative Analysis

Cookies are an indispensable part of the confectionery industry, highly valued for their taste and diversity. Comprehending the biochemical makeup of cookies is vital for optimizing recipes, enhancing nutritional content and ensuring product quality [26]. Proteins, carbohydrates and amino acids are some of the key components in the cookies, each playing a role in the texture, flavor and nutritional profile of the cookie. Proteins contribute to the structural integrity and texture of cookies. Carbohydrates are essential for sweetness and browning. Amino acids, the building of blocks of proteins [27], affect flavor and can provide insights into protein quality and interactions during baking. This evaluation utilizes qualitative tests to detect and analyze these macromolecules in cookies [28].

Xanthoprotein Test

A minute portion of the cookie specimen is immersed in concentrated nitric acid. The solution is then carefully warmed for a few minutes. The emergence of a yellow hue signifies the production of xanthoproteins, verifying the existence of proteins in the specimen [29,30].

Millon's Test

The cookie specimen is dissolved in pure water, followed by the addition of Millon's reagent (a mixture of mercury nitrate and nitric acid). The solution is then gently heated. The appearance of a red color indicates the presence of tyrosine. A positive result in the Millon's test confirms the existence of phenolic compounds, which play a role in enhancing flavor and promoting browning reactions during the baking process [29,30].

Ninhydrin Test

The test for ninhydrin is used to find proteins and free amino acids in the sample of cookies in which after dissolving a cookie sample in water, ninhydrin reagent is added. The mixture is warmed in a bath of water. The presence of free amino acids is indicated by a purple or blue hue. The presence of free amino acids, improves the flavor and nutritional value of the cookies, is suggested by a positive ninhydrin test [29,30].

Fehling's Test

Reducing sugars are crucial for the sweetness and caramelization of cookies, and this test finds them. For this test the cookie sample is combined with equal volumes of Fehling's B (sodium potassium tartrate) and Fehling's A (copper (II) sulfate) solutions. Heat is applied to the mixture. Reducing sugars are present when the precipitate turns from blue to brick-red in hue.
A positive Fehling's test indicates that reducing sugars are present, which affects the cookies' overall flavor profile and browning [29,30].

Benedict's Test

The Benedict's test aims to identify decreasing sugar levels in the cookie sample, much like Fehling's test did. For this test add Benedict's reagent after dissolving the cookie sample in water. In a bath of boiling water, the mixture is warmed. Reducing sugars are present when the color changes from blue to green, yellow, or red, with the strength of the color according to the sugar content [29,30].

GI index

The glycemic index is a measure of how different carbohydrate - containing foods affect blood glucose levels. A higher glycemic index value signifies a greater impact on blood sugar levels [31]. The amount that a food will increase a person's blood glucose levels after consumption is measured by its glycemic load (GL). The glycemic index (GI) of a food is determined by multiplying its GI by the quantity of carbs in a serving, and then dividing the result by 100. As shown in the diagram given below, compared to high-GI foods, low-GI foods are absorbed and digested more slowly.

Total Energy

Total energy was estimated on the basis of carbohydrate, protein, and fat content of cookie sample.

$$\text{Total energy} = \text{Carbohydrates} \times 4 + \text{Protein} \times 4 + \text{Fat} \times 9 \quad (1)$$

Protein Estimation

The DGHS Manual's prescribed method was employed for protein estimation. The procedure involved placing 200-300 mg of powdered cookies into four test tubes, followed by the addition of 3 g of catalyst (K_2SO_4+$CuSO_4$). Subsequently, 10 mL of concentrated H_2SO_4 was introduced to each tube, and the samples were digested for 3-4 hours. Post-digestion, the samples underwent distillation with boric acid, potassium permanganate, and 40% sodium hydroxide. The resulting solution was then titrated with acid to neutralize the ammonia. The protein percentage was determined using a specific equation based on this titration process.

$$\text{Ammonia \%} = \frac{14.01 \times (\text{Reading} - \text{Blank}) \times 100 \times \text{Normality}}{Sample\ Weight} \times 1000 \quad (2)$$

$$\text{Protein content} = \text{Ammonia \%} \times 6.25 \quad (3)$$

Carbohydrate Estimation

The carbohydrate content was measured using the DGHS Manual technique. To assess the carbohydrate levels, researchers placed 2 g of powdered cookies in a 200 ml volumetric flask and introduced 50 ml of lead acetate. They then added 6 ml of 0.5 N HCl and subjected the mixture to heat in a hot water bath. After heating sample were cooled and neutralized with 6 ml of 0.5 N NaOH; finally, the sample volume was made up to 200 ml by using distilled water. Invert sugar was determined before inversion by Lane and Eynon method. According to method 10 ml of mixed Fehling A and B solutions was taken in a conical flask and titration was carried out with sample solution within 3 min without inversion by using 1% aqueous Methylene Blue as an indicator.

$$\text{Reducing sugar \% before inversion} = \frac{F \times 10}{C \times R} \quad (4)$$

Where,
C = Concentration;
R = Reading;
F = Factor of Fehling solution

$$\text{Reducing sugar \% after inversion} = \frac{F \times 10}{C \times R} \quad (5)$$

Where,
C = Concentration;
R = Reading;
F = Factor of Fehling solution

$$\text{Total Carbohydrates \%} = \frac{\text{Total invert sugar after inversion}}{\text{Invert sugar \% before inversion}} \times 0.95 \quad (6)$$

Fat Estimation

The experimental method involved placing 2 g of the sample in a Soxhlet apparatus with a 1:1 mixture of diethyl alcohol and petroleum ether for 6 hours. Subsequently, the ether was eliminated through distillation, and the sample was dried in a hot air oven at 110 ± 1 °C before being cooled in a desiccator. The dried sample was then weighed again. The remaining residue underwent repeated washing with 2 to 3 mL of diethyl ether, and this process continued until a constant weight was achieved [32].

$$\text{\% of fat content} = \frac{(M1 - M2) \times 100}{\text{Weight of sample}} \quad (7)$$

Where,
M1 =Weigh of Round bottom flask with fat;
M2 = Weigh of the Round bottom flask

Results

Evaluation of Prepared Cookies

Sensory Evaluation Including Color, Odour, Taste, and Texture

The sensory evaluation demonstrated that the formulated cookies achieved good overall acceptability, with panelists providing favorable ratings across odor, taste, color, and texture on the 7-point hedonic scale. The mildly sweet and pleasant aroma suggested that the incorporation of green unripe jackfruit flour contributed positively to the aromatic profile, enhancing the perceived freshness and natural fragrance of the cookies. Taste evaluations indicated a slightly sweet flavor, which was well received and appropriate for a product intended for health-conscious consumers. The cookies exhibited a crunchy, coarse-crumb texture, a characteristic that was consistently appreciated by the panelists and contributed to a satisfying mouthfeel. The brownish appearance of the cookies also scored well, reflecting uniform Maillard browning during baking and contributing to their visual appeal. Collectively, these sensory attributes indicate that the formulation produced cookies with desirable organoleptic qualities, supporting their acceptability while aligning with the functional and nutritional goals of the product. These summarized sensory outcomes (Table 2) indicate that the formulation successfully produced cookies with appealing sensory qualities while retaining the desired nutritional attributes.

Table 2. Sensory evaluation of cookies

Sr./ No.	Parameter	Result
1	Odour	Sweetaromatic
2	Taste	Slightlysweet in taste
3	Colour	Brownish
4	Texture	Crunchy and coarsecrumb

Qualitative Chemical Analysis

The qualitative biochemical tests confirmed the presence of key macromolecules—proteins, amino acids, and reducing carbohydrates—which contribute significantly to the structural, sensory, and functional characteristics of the formulated cookies. Positive results in the Xanthoproteic and Millon's tests indicate the presence of proteins and phenolic amino acids such as tyrosine, which support structural integrity and promote Maillard browning, consistent with the brownish color observed during sensory evaluation. The Ninhydrin test further confirmed the presence of free amino acids, suggesting their role in flavor development through thermal interactions during baking. Both Fehling's and Benedict's tests yielded positive reactions for reducing sugars, indicating their contribution to sweetness and caramelization, which aligns with the slightly sweet taste and appealing browned appearance reported by panelists. Collectively, these biochemical findings validate the sensory attributes observed and demonstrate that the combination of unripe jackfruit flour and jowar provides an appropriate macronutrient profile to support desirable flavor, color, and texture. The consolidated results are presented in Table 3.

Table 3. Results of qualitative tests (Chemical analysis)

Sr./ No.	Test	Inference	Result
1	Xanthoprotein	Proteins	Positive
2	Millons	Proteins	Positive
3	Ninhydrin	Amino acids	Positive
4	Fehlings	Carbohydrates	Positive
5	Benedicts	Carbohydrates	Positive

GI Index Analysis

The glycemic index assessment provided insight into the potential postprandial glucose response associated with the formulated cookies. The medium GI values of the primary ingredients green unripe jackfruit flour and jowar indicate moderately slow carbohydrate digestion, which contributes to a more controlled release of glucose. Dates, included as a natural sweetener, demonstrated a low GI value, further supporting a reduced glycemic impact. The formulated cookies themselves exhibited a medium GI (50–55), suggesting that they are likely to produce a moderate, steady rise in blood glucose rather than a rapid spike. This response is consistent with the combination of complex carbohydrates and natural sugars present in the formulation. Although the cookies fall within the medium GI category, their glycemic characteristics align with expectations for products containing whole grain and fruit-based ingredients. The GI values for raw ingredients and the final product are summarized in Table 4.

Table 4. Glycemic index chart

Sr./ No.	Ingredient	GI range	GI value
1	Green unripe jackfruit	50-60	Medium
2	Jowar	50-62	Medium
3	Dates	40-42	Low
4	Cookies	50-55	Medium

Proximate Analysis

The proximate composition analysis provided insight into the nutritional profile of the formulated cookies. The cookies exhibited a high carbohydrate content (67.2 g/100 g), which is typical of cereal-based baked products and contributes substantially to the total energy value of 525 kcal/100 g. The protein content (5.63 g/100 g) reflects the contributions of jackfruit flour and jowar, supporting structural stability and browning reactions noted in sensory and qualitative assessments. The fat content (26 g/100 g) is consistent with standard cookie formulations and contributes to the desirable crispness and mouthfeel observed during sensory evaluation. However, the total sugar content (37.6 g/100 g), including 32 g/100 g added sugar, is relatively high and represents a nutritional limitation, particularly for individuals with diabetes or those seeking low-sugar dietary options. Despite the moderate glycemic index observed, the elevated sugar content reduces the suitability of the product as a diabetic-friendly food. These findings, summarized in Table 5, highlight both the functional advantages and nutritional constraints of the cookie formulation.

Table 5. Proximate composition of cookies

Sr./ No.	Nutritional Parameters	Result
1	Energy value	525 kcal/100g
2	Protein	5.63 g/100g
3	Carbohydrates	67.2 g/100g
4	Sugar(Total)	37.6 g/100g
5	Fat	26 g/100g
6	Added sugar	32 g/100g

The nutritional analysis presented in Table 5 indicates the formulation shows a comparatively lower glycemic index, the total sugar content (32 g/100 g) restricts its suitability for individuals with diabetes and therefore the product should not be classified as diabetic-friendly.

Discussion

Sensory Evaluation

The sensory evaluation demonstrated that the green, unripe jackfruit–based cookies were generally well accepted by the panelists, indicating good potential for consumer acceptability. The mildly sweet and pleasant

aroma suggests that the natural volatile compounds present in jackfruit flour contributed positively to the fragrance profile. Panelists also appreciated the slightly sweet flavor, which aligns with the intended health-oriented formulation by avoiding excessive sweetness. The cookies exhibited a crunchy, firm texture that was consistently rated favorably, reflecting the functional contribution of jackfruit flour and jowar to the product's structural properties. The brownish appearance was also well received, correlating with desirable Maillard browning reactions. Collectively, these findings indicate that the formulation successfully produced cookies with appealing sensory characteristics suitable for health-conscious consumers.

Qualitative Analysis of Macromolecules

The qualitative biochemical tests provided insight into the macromolecular components that influence the functional and sensory properties of the cookies. Positive Xanthoproteic and Millon's tests confirmed the presence of proteins and phenolic amino acids, which are known to enhance flavor and contribute to color development through Maillard reactions during baking. The Ninhydrin test further validated the presence of free amino acids, supporting their role in flavor enhancement and nutritional contribution. The detection of reducing sugars through Fehling's and Benedict's tests reinforces their significant role in caramelization and browning, which directly relates to the desirable appearance and mildly sweet taste noted in sensory evaluation. These findings confirm that the biochemical composition of the cookies aligns with and supports the observed sensory attributes

Glycemic Index and Glycemic Load

The glycemic index assessment indicated that the formulated cookies fall within the medium GI range (50–55), suggesting a moderate postprandial glucose response. This is consistent with the carbohydrate sources used in the formulation, including green unripe jackfruit flour and jowar, both of which possess moderate GI values. The presence of dates, a low-GI ingredient, further supports the moderation of glycemic response. Glycemic load calculations provided a more comprehensive understanding by incorporating both GI and carbohydrate quantity, indicating that the cookies produce a slower and more sustained rise in blood glucose compared with high-GI foods. These characteristics make the cookies a potentially suitable option for individuals seeking controlled glucose responses, though mindful consumption is still required due to overall carbohydrate and sugar content.

Proximate Composition

The proximate composition results confirmed that the cookies contain substantial carbohydrates (67.2 g/100 g), moderate protein content (5.63 g/100 g), and a relatively high fat level (26 g/100 g), contributing to an overall energy value of 525 kcal/100 g. These nutrient levels are typical of bakery products, and the moderately high fat content likely contributed to the desirable crispness and mouthfeel observed during sensory evaluation. However, the total sugar content—particularly the added sugar component (32 g/100 g)—presents limitations for diabetic individuals despite the medium GI classification. While the cookies exhibit functional benefits from jackfruit flour and jowar, the elevated sugar levels indicate that the product should not be strictly classified as diabetic-friendly. Instead, it may be better positioned as a healthier alternative within general consumer markets rather than a specialized diabetic product.

Limitations of this Study

This study is limited by the absence of clinical parameters such as HbA1c, fasting blood glucose, or postprandial glycemic response, which restricts the ability to determine the true anti-diabetic impact of the formulated cookies. Additionally, the glycemic index (GI) estimation was based on laboratory-derived values of individual ingredients rather than direct *in vivo* or standardized *in vitro* GI testing of the final product. Therefore, the reported GI should be interpreted cautiously, and future studies should include clinical validation and standardized GI measurement protocols.

Conclusion

Bakery items have gained considerable popularity in a recent times, catering to individuals from diverse backgrounds. Among these, cookies are the most widely consumed and often provide an opportunity to incorporate nutrients and nutraceutical components that promote health. The use of nutritional ingredients in cookies results in improved visual appeal, texture, taste, and overall acceptability. Additionally, these cookies exhibit a lower glycemic impact compared to traditional ones, making them a more desirable option. The inclusion of green unripe jackfruit in the formulation of cookies results in a low glycemic index, which could potentially provide health benefits for individuals suffering from degenerative diseases such as diabetes mellitus. The incorporation of functional ingredients like green unripe jackfruit, dates and sorghum in the development of novel products holds great promise for the food industry. Thus, incorporating green unripe jackfruit flour with sorghum flour shows significant promise for creating anti-diabetic cookies. This approach not only helps reduce waste but also presents opportunities for cost reduction in the production of baked goods. The use of unripe jackfruit in cookie manufacturing addresses both environmental concerns and economic efficiency in the baking industry.

ORCID

Anjali D. Kshirsagar
https://orcid.org/0009-0002-2318-0836
Tanishk D. Khairnar
https://orcid.org/0009-0000-4403-2467
Payal V. Patil
https://orcid.org/0009-0001-8422-7290
Rakesh U. Shelke
https://orcid.org/0000-0002-3822-4558
Eknath D. Ahire
https://orcid.org/0000-0001-6542-884X
Khemchand R. Surana
https://orcid.org/0000-0001-8918-1159

References

[1] Antar, S.A., Ashour, N.A., Sharaky, M., Khattab, M., Ashour, N.A., Zaid, R.T., Al-Karmalawy, A.A. Diabetes mellitus: Classification, mediators, and complications; A gate to identify potential targets for the development of new effective treatments. *Biomedicine & Pharmacotherapy*, **2023**, 168, 115734.
[2] Colagiuri, S. Definition and classification of diabetes and prediabetes and emerging data on phenotypes. *Endocrinology and Metabolism Clinics*, **2021**, 50(3), 319-336.
[3] Duan, D., Kengne, A.P., Echouffo-Tcheugui, J.B. Screening for diabetes and prediabetes. *Endocrinology and Metabolism Clinics*, **2021**, 50(3), 369-385.
[4] Rohit, G., Vishnu, M., Anjali, S. Herbal remedies for type 2 diabetes mellitus: An explicit review. *Current Drug Therapy*, **2025**, 20(7), 1034-1052.
[5] Palamthodi, S., Shimpi, S., Tungare, K. A study on nutritional composition and functional properties of wheat, ragi and jackfruit seed composite flour. *Food Science and Applied Biotechnology*, **2021**, 4(1), 63-75.
[6] Menaka, U., Wijesekara, I., Navaratne, S., Wickramasinghe, I. Gluten-free bakery foods enriched with plant proteins: An overview. *Food Reviews International*, **2024**, 40(10), 3847-3868.
[7] Aschemann-Witzel, J., Gantriis, R.F., Fraga, P., Perez-Cueto, F.J. Plant-based food and protein trend from a business perspective: Markets, consumers, and the challenges and opportunities in the future. *Critical Reviews in Food Science and Nutrition*, **2021**, 61(18), 3119-3128.
[8] Tan, M., Nawaz, M.A., Buckow, R. Functional and food application of plant proteins–a review. *Food Reviews International*, **2023**, 39(5), 2428-2456.
[9] Olokoba, A.B., Obateru, O.A., Olokoba, L.B. Type 2 diabetes mellitus: A review of current trends. *Oman Medical Journal*, **2012**, 27(4), 269.
[10] Kurtzhals, P., Kreiner, F.F., Bindra, R.S. The role of weight control in the management of type 2 diabetes mellitus: Perspectives on semaglutide. *Diabetes Research and Clinical Practice*, **2023**, 203, 110881.
[11] Müller, T.D., Finan, B., Bloom, S., D'Alessio, D., Drucker, D.J., Flatt, P., Habener, J. Glucagon-like peptide 1 (GLP-1). *Molecular Metabolism*, **2019**, 30(72-130.
[12] Mariam, Z., Niazi, S.K. Glucagon-like peptide agonists: A prospective review. *Endocrinology, Diabetes & Metabolism*, **2024**, 7(1), e462.
[13] Rasmussen, M.F. The development of oral semaglutide, an oral GLP-1 analog, for the treatment of type 2 diabetes. *Diabetology international*, **2020**, 11(2), 76-86.

[14] Demirkesen, I., Ozkaya, B. Recent strategies for tackling the problems in gluten-free diet and products. *Critical Reviews in Food Science and Nutrition*, **2022**, 62(3), 571-597.
[15] Rao, A.G., Naik, K.S., Unnikrishnan, A., Joseph, J. Efficacy of green jackfruit flour as a medical nutrition therapy replacing rice or wheat in patients with type 2 diabetes mellitus: A randomized, double-blind, placebo-controlled study. *Nutrition & Diabetes*, **2021**, 11(1), 18.
[16] Ye, S., Imran, A., Agar, O.T., Yadav, D., Moore, C., Suleria, H.A. Jackfruit (*Artocarpus heterophyllus* Lam.): Nutritional profile, polysaccharide analysis, and opportunities for product development. *International Journal of Food Science and Technology*, **2025**, 60(1), 91.
[17] Seino, Y., Sasaki, T., Fukatsu, A., Ubukata, M., Sakai, S., Samukawa, Y. Efficacy and safety of luseogliflozin as monotherapy in Japanese patients with type 2 diabetes mellitus: A randomized, double-blind, placebo-controlled, phase 3 study. *Current Medical Research and Opinion*, **2014**, 30(7), 1245-1255.
[18] Konsue, N., Bunyameen, N., Donlao, N. Utilization of young jackfruit (*Artocarpus heterophyllus* Lam.) as a plant-based food ingredient: Influence of maturity on chemical attributes and changes during *in vitro* digestion. *LebensonWissTechnol,* **2023,**180, 114721.
[19] Rana, S.S., Pradhan, R.C., Mishra, S. Image analysis to quantify the browning in fresh cut tender jackfruit slices. *Food Chemistry*, **2019**, 278, 185-189.
[20] Goel, B., Nanda, S. Development and sensory evaluation of a value added food product for prevention of diabetes. *International Journal of Health Sciences*, II), 10303-10307.
[21] Chiranthika, N., Chandrasekara, A., Gunathilake, K. Physicochemical, in vitro antidiabetic and sensory characteristics of leavened functional bread made with Lasia spinosa and Nelumbo nucifera rhizome flours composited with wheat flour. *Food Production, Processing and Nutrition*, **2024**, 6(1), 33.
[22] Arise, A.K., Malomo, S.A., Acho, M.A., Ajao-Azeez, N.D., Arise, R.O. *In vivo* anti-diabetic activity, physicochemical and sensory properties of Kunu enriched with African walnut. *Food Chemistry Advances*, **2023**, 2, 100315.
[23] Goh, S.X., Mamat, H., Aziz, A.H.A. Valorization of agriculture by-product: Development of gluten-free biscuit made from blends of okara and jackfruit seed flour. *Waste Management Bulletin*, **2024**, 2(2), 59-65.
[24] Sayem, A., Talukder, S., Akter, S.S., Alam, M., Rana, M.R., Alam, M.M. Utilization of fruits and vegetables wastes for the dietary fiber enrichment of biscuits and its quality attributes. *Journal of Agriculture and Food Research*, **2024**, 15, 101077.
[25] Ortega-González, L., Güemes-Vera, N., Piloni-Martini, J., Quintero-Lira, A., Soto-Simental, S. Substitution of wheat flour by jackfruit (*Artocarpus heterophyllus* lam.) seed flour: Effects on dough rheology and deep-frying doughnuts texture and sensory analysis. *International Journal of Gastronomy and food Science*, **2022**, 30, 100612.
[26] Hettiaratchi, U., Ekanayake, S., Welihinda, J. Nutritional assessment of a jackfruit (*Artocarpus heterophyllus*) meal. *Ceylon Medical Journal*, **2011**, 56(2).
[27] Ahiduzzaman, M., Islam, M.N., Jannati, S.F.T., Ali, M.A., Hossain, M.M. Preserving jackfruit: unlocking year-round availability as industrial ingredient for value-added food products. *Journal of Agriculture and Food Research*, **2024**, 16, 101184.
[28] Li, Z., Lan, Y., Miao, J., Chen, X., Chen, B., Liu, G., Cao, Y. Phytochemicals, antioxidant capacity and cytoprotective effects of jackfruit (*Artocarpus heterophyllus* Lam.) axis extracts on HepG2 cells. *Food Bioscience*, **2021**, 41, 100933.
[29] Tao, Y.M., Wang, S., Luo, H.L., Yan, W.W. Peroxidase from jackfruit: Purification, characterization and thermal inactivation. *International Journal of Biological Macromolecules*, **2018**, 114, 898-905.
[30] Kaur, R., Bobade, H., Sachdev, P.A., Kaur, S. Jackfruit kernel starch-composition, structure, properties and modifications. In *Non-Conventional*

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

[31] Ngo, H.B.G., Phu, M.L., Tran, T.T.T., Ton, N.M.N., Nguyen, T.Q.N., Le, V.V.M. Dietary fiber-and antioxidant-enriched cookies prepared by using jackfruit rind powder and ascorbic acid. Heliyon, 2024, 10(9), 30884.

[32] Khan, A.U., Ema, I.J., Faruk, M.R., Tarapder, S.A., Khan, A.U., Noreen, S., Adnan, M. A review on importance of Artocarpus heterophyllus L. (Jackfruit). Journal of Multidisciplinary Applied Natural Science, 2021, 1(2), 106-116.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.